K-Nearest Neighbour (K-NN)

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OUTLINE

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- Other Distance Metrics
- K-NN Classifier
- Decision Boundary of K-NN Classifier
- Choosing the value of K
- Merits and Demerits of K-NN classifier
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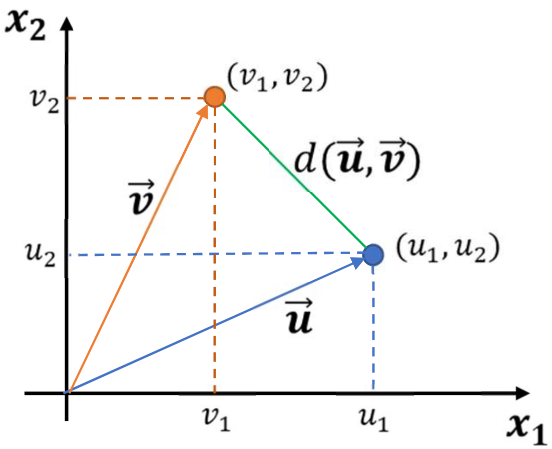
EUCLIDEAN DISTANCE



- Consider the points in two-dimension. Each point in two-dimension can be represented by a vector of dimension two.
- The point (u_1, u_2) can be represented by the vector $\vec{\boldsymbol{u}} = [u_1, u_2\ ]^T$
- And the point (v_1, v_2) can be represented by the vector $\vec{\boldsymbol{v}} = [v_1, v_2\ ]^T$
- The Euclidean distance between the points is:

$$d(\vec{\boldsymbol{u}}, \vec{\boldsymbol{v}}) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}$$

- In general a point in n-dimensional space is represented as $\vec{\boldsymbol{u}} = [u_1, u_2, u_3, \ldots, u_n\ ]^T$, a n-D vector
- Hence, the Euclidean distance between two points in n-dimensional space is represented as:

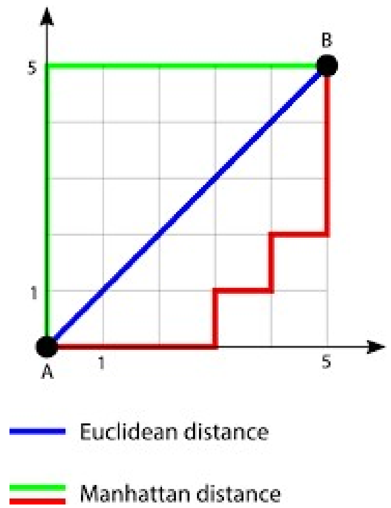
$$d(\vec{\boldsymbol{u}}, \vec{\boldsymbol{v}}) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2 + \ldots + (u_n - v_n)^2} = \sqrt{\sum_{i=1}^{n} (u_i - v_i)^2}$$

- In general in vector notation, the Euclidean distance is written as: $d(\vec{\boldsymbol{u}}, \vec{\boldsymbol{v}}) = \sqrt{(\vec{\boldsymbol{u}} - \vec{\boldsymbol{v}})^T (\vec{\boldsymbol{u}} - \vec{\boldsymbol{v}})}$

OTHER DISTANCE METRICS

- **Manhattan Distance:** For two data points denoted by $\boldsymbol{x}$ and $\boldsymbol{y}$ the Manhattan distance is defined as:

$$dist_{manhattan}(\boldsymbol{x},\boldsymbol{y}) = \sum_{i=1}^{n} |x_i - y_i|$$



- **Minkowski Distance:** For two data points denoted by $\boldsymbol{x}$ and $\boldsymbol{y}$ the Minkowski distance is defined as:

$$dist_{minkowski}\,(\boldsymbol{x},\boldsymbol{y},h) = [\textstyle\sum_{i=1}^{n}(x_i - y_i)^h]^{\left(\frac{1}{h}\right)}$$

Note: for $h = 2$, Minkowski Distance is same as Euclidean Distance and for $h = 1$, it is Manhattan Distance

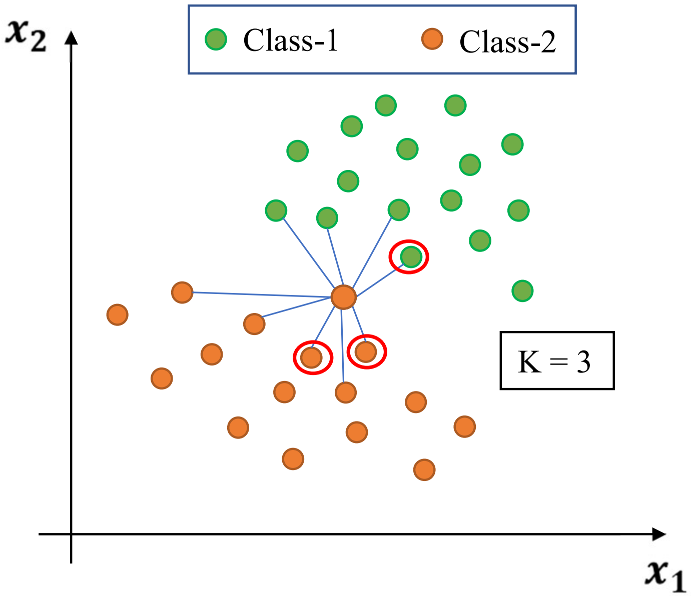
- **Chebyshev Distance:** For two data points in n-dimensional space it is defined as:

$$dist_{chebyshev}\,(\boldsymbol{x},\boldsymbol{y}) = \max(|x_1 - y_1|,\ |x_2 - y_2|, |x_3 - y_3|, \dots, |x_n - y_n|)$$

There are other distance metric like: [**Mahalanobis** Distance], [**Bhattacharyya** Distance] etc. which are used for advanced statistical pattern recognition tasks.

K-NN CLASSIFIER

- **Intuition:** One is known by the company one keeps.

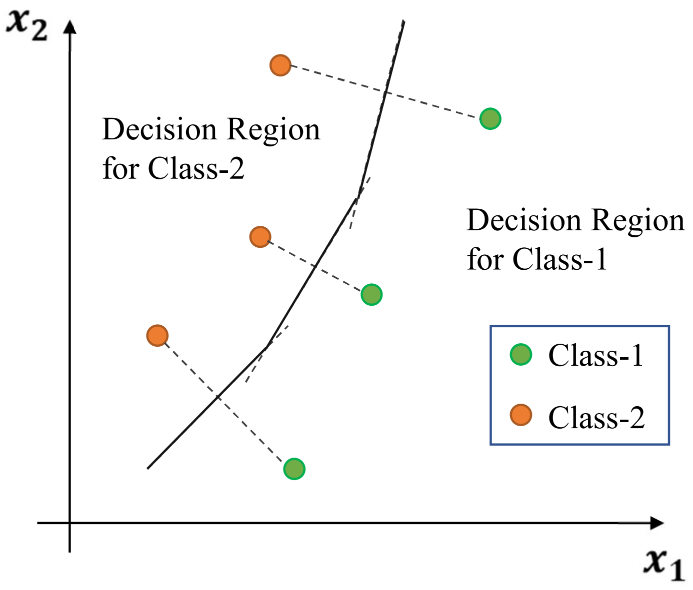


- **K-NN Algorithm:**
  1. All the training samples/ points are available beforehand.
  2. When a new test sample arrives calculate its **distance** from **all training points.**
  3. Choose K-nearest neighbours based on the distance calculated. Usually the K is a positive odd integer and supplied by user.
  4. Assign the class label of the test sample **based on majority**. i.e. for a test sample if most number of neighbours among those K-Nearest Neighbours belong to one particular class-c, then assign the class label of test sample as c.

- **Characteristics of K-NN Classifier:**

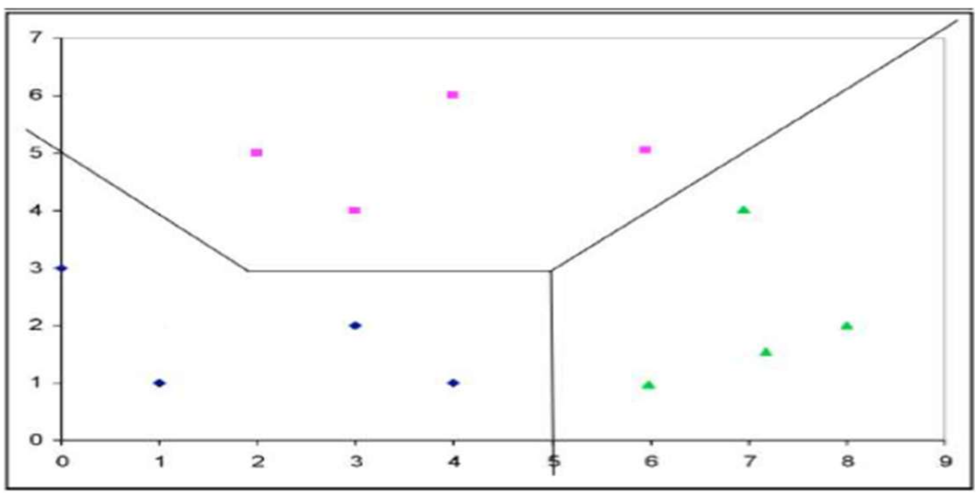
It doesn't create model based on the training patterns in advance. Rather, when a test instance comes for testing, runs the algorithm to get the class prediction of that particular testing instance. Hence, there is no learning in advance. Hence, k-NN classifier is also known as **Lazy Learner**. Because KNN relies on distance metrics, features with larger scales can dominate the distance calculation. It is mandatory to scale or normalize features.

K-NN CLASSIFIER: DECISION BOUNDARY



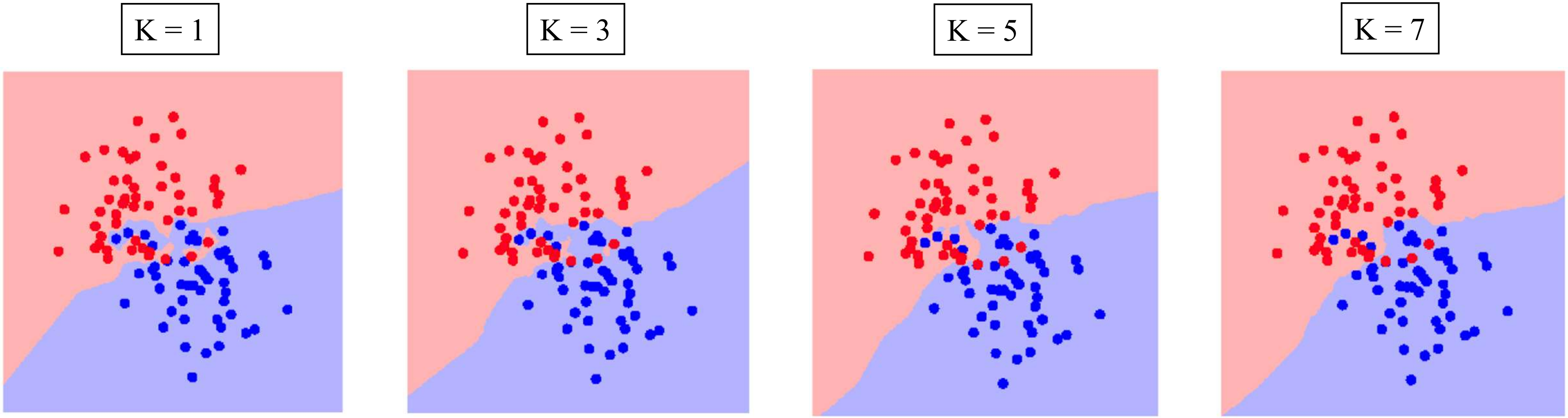
- Boundary are the points those are equidistant between the points of Class-1 and Class-2
- Construct lines between closest pairs of points in different classes.
- Draw perpendicular bisectors. End bisectors at intersections.
- Note that locally the boundary is linear.
- Hence the decision boundary is piecewise linear curve.

For multiclass classification also the same thing is done to find the decision boundary.



K-NN: CHOOSING THE VALUE OF K

- Increasing the 'K' simplifies the decision boundary. Because majority voting implies less emphasis on individual points



- However increasing the K also increases computational cost.
- Hence, choosing K is an optimization between how much simplified decision boundary we want vs. how much computational cost we can afford.
- Usually K = 5, 7, 9, 11 works fine for most practical problems.

K-NN CLASSIFIER: MERITS AND DEMERITS

Merits:

- K-NN Classifier often works very well for practical problems.
- It is very easy to implement, as there is no complex learning algorithm involved.
- Robust to Noisy Data.

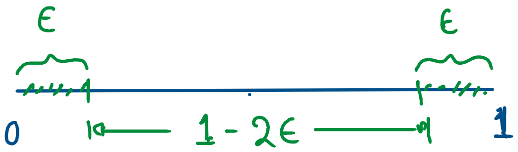
Demerits:

- Choosing the value of K may not be straightforward. Often the same training samples are used for different values of K, and we choose the most suitable value of K based on minimum misclassification errors on test samples.
- Doesn't work well for categorical attributes.
- Can encounter problem with sparse training data. (i.e. data points are located far away from each other)
- Can encounter problems in very high-dimensional spaces.
    - Most points are at corners.
    - Most points are at the edge of the space.

  This problem is known as ***Curse of Dimensionality*** and affect many other Machine Learning algorithms.

Understanding Curse of Dimensionality

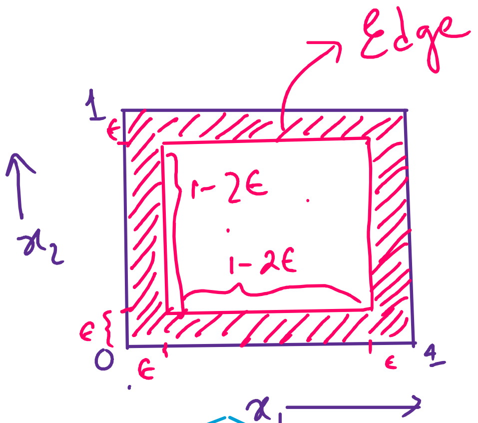
1-D



Edge is ϵ distance away from either end.

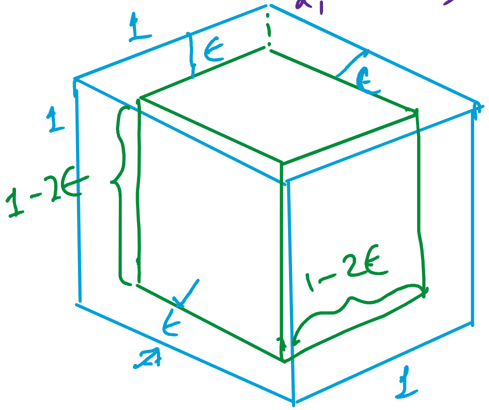
$P(\text{Edge}) = 2\epsilon$ $\qquad$ $P(\text{not Edge}) = 1-2\epsilon$

2-D



$P(\text{edge}) = 1-(1-2\epsilon)^2$ $\qquad$ $P(\text{not edge}) = (1-2\epsilon)^2$

3-D



$P(\text{Edge}) = 1-(1-2\epsilon)^3$ $\qquad$ $P(\text{not Edge}) = (1-2\epsilon)^3$

For n-dimensional hyper-cube

$$\therefore P(\text{Edge}) = 1-(1-2\epsilon)^n$$

$\boxed{\epsilon} \rightarrow 0.05$

$1-2\epsilon = 0.9$

$P(\text{Edge}) = 1-(0.9)^n$

$n = 50$

$\therefore \underline{P(\text{Edge})} = 1 - 0.0052$

$= 0.9948$

K-NN REGRESSION

K-nearest neighbors (KNN) regression is a **non-parametric supervised machine learning method** that predicts the continuous value of a new data point by averaging the target values of its 'K' nearest neighbors in the training dataset.

How it works?

The process for a K-NN regressor involves a few key steps for each new data point needing a prediction:

1. **Select '*K*':** Choose the number of neighbors (K) to consider. The optimal K is typically determined using cross-validation to balance bias and variance trade-offs.

2. **Calculate Distances:** Compute the distance (e.g., Euclidean distance, Manhattan distance, Minkowski distance) between the new data point and all existing points in the training dataset.

3. **Identify Neighbors:** Select the K training data points that are closest to the new point.

4. **Predict:** The predicted output value is the mean (or a weighted average, giving more influence to closer neighbors) of the target values of these K neighbors.

The merits and Demerits of K-NN Regressor are same as that of K-NN classifier.

Thank You